Analysis of Data from the Small Craft ATR (SCATR) Program

Including an Algorithm for Extrapolating Data from a Fixed Radar to a Moving Radar at a Different Range

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The purpose of this note is to document the work I have done on my new algorithm to extrapolate shore radar data to a moving platform at a different range and to read out my current analysis of the Tybee results from SCATR. To simplify things, I have just included a zip file for the current working directory. This file covers the GPS data, the raw data, and the Matlab codes for the work.

I have chosen to extrapolate the Tybee data by a factor of 10 in range, from less than 9 km to less than 90 km. I added a cross-range platform motion of 50 m/s. This motion is equivalent to flying toward the target at 100 m/s and then turning 30 degrees for the ISAR collection.

To see the figures that the publication can draw from, just run the two Matlab codes TybeeGPS and TybeeIAT. One does not need to execute these codes since the published directory has all the results for both codes. The user is of course allowed to modify the parameters or the code output to understand the algorithm.

TybeeGPS reads the GPS data file and produces the background information on the range, velocity, heading and aspect angles that TybeeIAT reads.

TybeeIAT(option) executes the IAT algorithm to produce the estimates of aspect angle and a variety of diagnostic plots. It stores the critical plots for each option in a directory named after the base integration time. For example, the output for the 240-second integration time (used in the paper) is in the directory 240T.

More details of the algorithms are summarized below.

Task 1: Analysis of the GPS data from SCATR

The first stage of this task is to use Excel to load and modify the CSV-file from NRL (SC15GPS.CSV) that contains GPS data and derived quantities sampled at a 1-second interval. I deleted the first 1879 rows of the sheet to skip the 4 header lines and 1875 samples (31.25 minutes) for the time the Tybee was leaving the harbor and sailing to the experiment site. Then I deleted the bottom rows after the ship left the site, leaving 7200 lines (2 hours) of active data. Then I deleted 3 columns containing the date, the time, and one that is all zeros. That leaves 8 columns of latitude and longitude plus derived quantities like range,

heading, and aspect; and estimates of the antenna azimuth and elevation. The array of data consisting of 8 columns by 7200 rows is then saved to a text file for input to Matlab.

I then wrote a Matlab program to read in this data and to analyze it. This code, TybeeGPS.m, reads the spreadsheet and plots some comparisons of my analysis with the NRL interpretations in the file. Finally, the important results of this analysis are output to a Matlab data file for input to Task 2 and following tasks. For this output I only used 3000 (50 minutes) of the two-hour data set since this covers all the time sequences when I have radar data for the Tybee.

Note: Tasks 2 through 5 are done in a separate Matlab code, TybeeIAT.m, which reads the auxiliary data and ISAR results for each of the 8 Tybee cases and produces advanced analysis of them.

Task 2: Processing of the Adaptive Motion Compensation from the ISAR Processor

In my ISAR processor the adaptive motion compensation data is stored in a text file (.moc) listing time, range offset and radial velocity. This file is read by the Matlab code and the first 32 pulses are skipped since they are used for the ISAR acquisition mode and represent redundant information. This data is then processed through the IAT algorithm for Data Preparation, which consists of median smoothing, lowpass smoothing and an adaptive narrow band smoother to remove the dominant ocean waves. After the final stage of smoothing the code does a linear regression process that breaks the time series into the mean, the linear trend and the remainder, which is labeled Noise. The algorithm for extrapolating this data to a different range or platform motion has options for including the mean value and fraction (Nmult) of the noise term and the linear trend (Amult).

The final three tasks use the GPS data and the ISAR motion compensation data to evaluate the cross-range velocity component and the aspect angle for three scenarios I call Model, Demo, and Data. These scenarios are all generated by the same subroutine (IAT_Test.m) using a different call sequence for each. IAT_Test calculates the range and range-rate versus time from the estimated ship and aircraft motion while adding the effects of the motion compensation noise and the acceleration error.

Task 3: ISAR AutoTrack Evaluation Assuming only the GPS Data (The Model Option)

In the Model option I ignore the radar data and calculate the Map-Drift algorithm using only the GPS data with no attempt to extrapolate to a different range or platform speed.

Task 4: ISAR AutoTrack with Noise-free ISAR Mocomp Data (The Demo Option)

In the Demo option I use both the GPS data and the ISAR data and I implement the extrapolation, but I ignore the noise. This process results in near perfect agreement with the

Map-Drift equation and a nearly exact solution to the cross-range velocity and the aspect angle.

Task 5: ISAR AutoTrack with ISAR Mocomp Including Noise (The Data Option)

In the Data option I use both the GPS data and the ISAR data and I implement the extrapolation, and I include both the quasi-random noise plus the errors in the acceleration.

What and where was the NRL GPS? Was it carried on the ships or was it a base station?

Are you sure that this was a differential GPS? If so then was the GPS spreadsheet produced after the test? Note that the GPS DATA does not have waves ... makes sense since the wave displacements were only 0.3 m.

I guess I would like to know the accuracy of the GPS. It looks pretty accurate since its acceleration agrees with the ISAR. But it does not see the waves. These facts argue for a differential GPS with accuracy of a few meters but not as good as one foot. I think that regular, non-differential GPS would have been worse.

According to Jay Trischman, the NRL GPS was in a big watertight case. There were two identical (or similar) GPU units that NRL brought. They put one on the ship and one at the radar field house. Each recorded the data. Then the differential calculations were done offline.

The two NRL GPS units now makes the story a lot cleaner. It confirms that SCATR used a differential GPS system. And it explains how they NRL GPS spreadsheet has the wrong direction of east or north motion, with the lat/lon listings showing motion from the harbor to the east, across a few major highways and then heading south to Mexico. This must be because they swapped the data from the primary and base systems.

Refer to my earlier email 'Tybee Results for Mocomp Acceleration'. I said that the first plot, comparing the GPS and ISAR acceleration estimates for the 32 60-second data segments. I said the two are correlated. Specifically, the correlation coefficient is 0.87. But the correlation coefficients of both these variables with the True Acceleration calculated from the GPS positions are 0.32 and 0.46. In other words, the acceleration estimates for GPS and ISAR data agree strongly but neither one agrees with the true value. Also note that both the GPS and ISAR accelerations contain about 1/3 negative values which should not happen if the ship heading is constant.

My conclusion from this info is that the GPS and ISAR data agree because they are both correctly measuring the actual ship motion. This means that the ship has a randomly varying heading with a magnitude of a degree or two. This is sufficient to cause a bend of the path that is larger than the effective bend due to the fact that a constant heading path has a small

variation in aspect angle due to the cross-range motion of the ship (i.e., the cross-range motion yields a change in bearing … and aspect angle is just the difference between the heading and the bearing).

Now, it is important to understand that the differential GPS data may be very accurate but that this information was not available to the helmsman of the Tybee to help it steer a perfectly straight course. Instead, the GPS output is only computed well after the experiment when the NRL people process the data recorded by the two units from the ship and the shore site. Also, note that there are no visual clues out there to steer to. One could have set buoys to define the octagon but they did not do this – they just gave a script to the Tybee to start at one point and then to execute a series of straight segments at given headings, speeds, and time durations.

Thus, when I plot the first 50 minutes (after the transit time to the site) of the Tybee GPS file I get the first plot above. This covers the cases sc15a07-sc15a14. But when I plot the first 120 minutes, I get the second plot where there are two circuits of the octagon in the same direction plus another half circuit in the opposite direction.

Raghu – Jay said that they never had the ships active without running the radar. This implies that there were originally more files starting at sc15a15. Does NRL have this data in your archives? I think that SPAWAR deleted their SCATR archives to save disk space (when that mattered!). Now it seems that I am the only person in the world who even has the full raw data for the 8 Tybee cases from March 6, 1998. The full data is in the form of '.nr0' files, each of which has over 5.5 minutes of data with the full date/time stamps and other auxiliary data.

I now have a good idea of the steering issues. My gut feeling is that without reference buoys the helmsman would not be able to steer a straight course for a 1600 m leg and not deviate a few m. I am convinced that this is the main explanation that the GPS and ISAR data are correlated and have similar variances. The wiggles in this path average out as we increase the dwell time … for both methods. And with a long enough dwell they both agree with the Map-Drift truth. (Acc = U-squared/R)

I think once the initial start point was established, the legs were
determined by heading and time.

I am trying to point to the development of a general model of ship length accuracy bases on the IAT and other concepts. A key measurement is the mocomp acceleration.

Jay helped me understand the SCATR data collection. He first noticed that the paths given by the GPS files were wrong but in an interesting and reversible way. Apparently, they were using a differential GPS with a base station at the radar site to compare with the GPS

recordings on the ships. But the analyst at NRL must have flipped the two terms and got the opposite direction of the bearing and path. So, I just flipped the change rates for East and North and then everything straightened out. Interestingly, the NRL analyst must have done this also since all the heading, aspect and azimuth values in the GPS spreadsheet agree with my values.

Jay also told me that they usually swapped the output file at each turn of the octagonal path. This clue allowed me to properly interpret the times from the 'small spreadsheet'. Apparently, the times are the start times of the files at each turn but not the times for the NRL data files. The data files are 60-second snippets that start after the ship has passed the turning point and has established a steady course. From the comparison of the GPS and ISAR data I estimated this lag to by about 50 seconds. However, there is some wiggle room in this estimate since I cannot see the waves fluctuations in the GPS data to sync them with the ISAR data. However, I had the raw (.nr0) files from Jay's radar and they are about 6 minutes long. Therefore, I can start the ISAR processing at the initial pulse of the file and continue to the end. When I compared this mocomp file to the file from the NRL 60-second snippets I found that the lag is exactly 1 minute, not 50 seconds. If you lag by 60 seconds the two mocomp curves almost completely overlap.

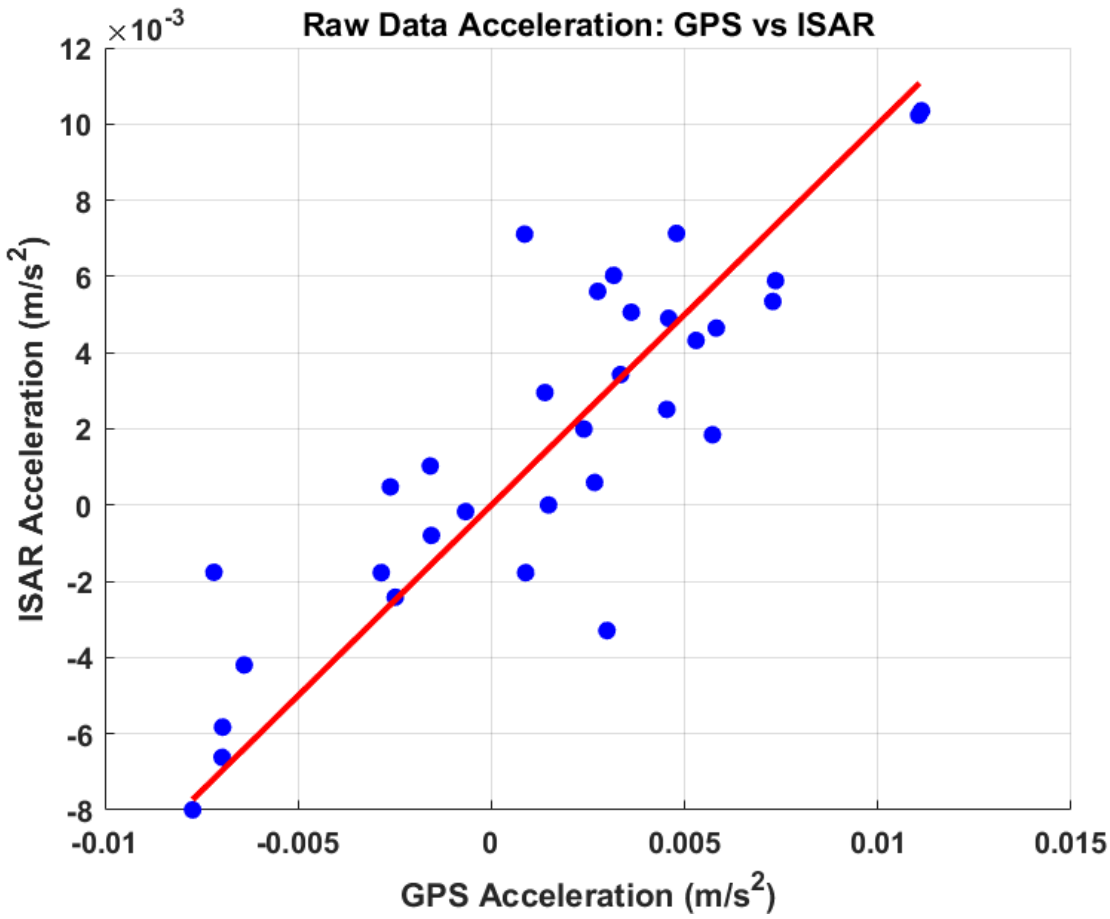
Since I have the original raw data for all 8 sides of the octagon for the Tybee data from March 6, 1998, I can break it up into 4 different 1-minute snippets per octagon side (skipping the first minute of course adjustment and the last fraction of a minute). Thus, instead of 8 cases at a dwell of 60 seconds I have 32 cases. Moreover, I can use longer dwell times for the IAT algorithm. So, I calculated the results for 32 60-second dwells, 16 120-second dwells and 8 240-second dwells.

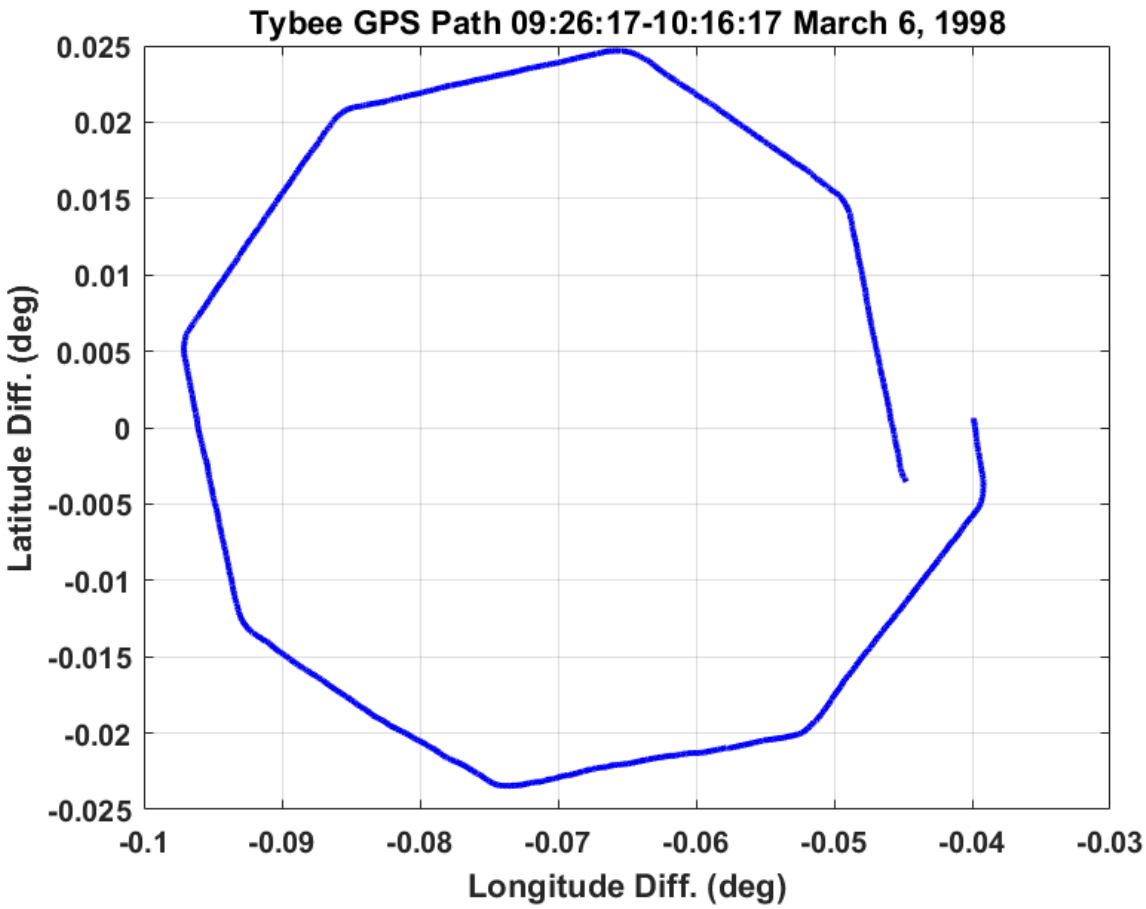
The first plot above is the acceleration estimates for the GPS data versus the ISAR processor data. Note 3 things:

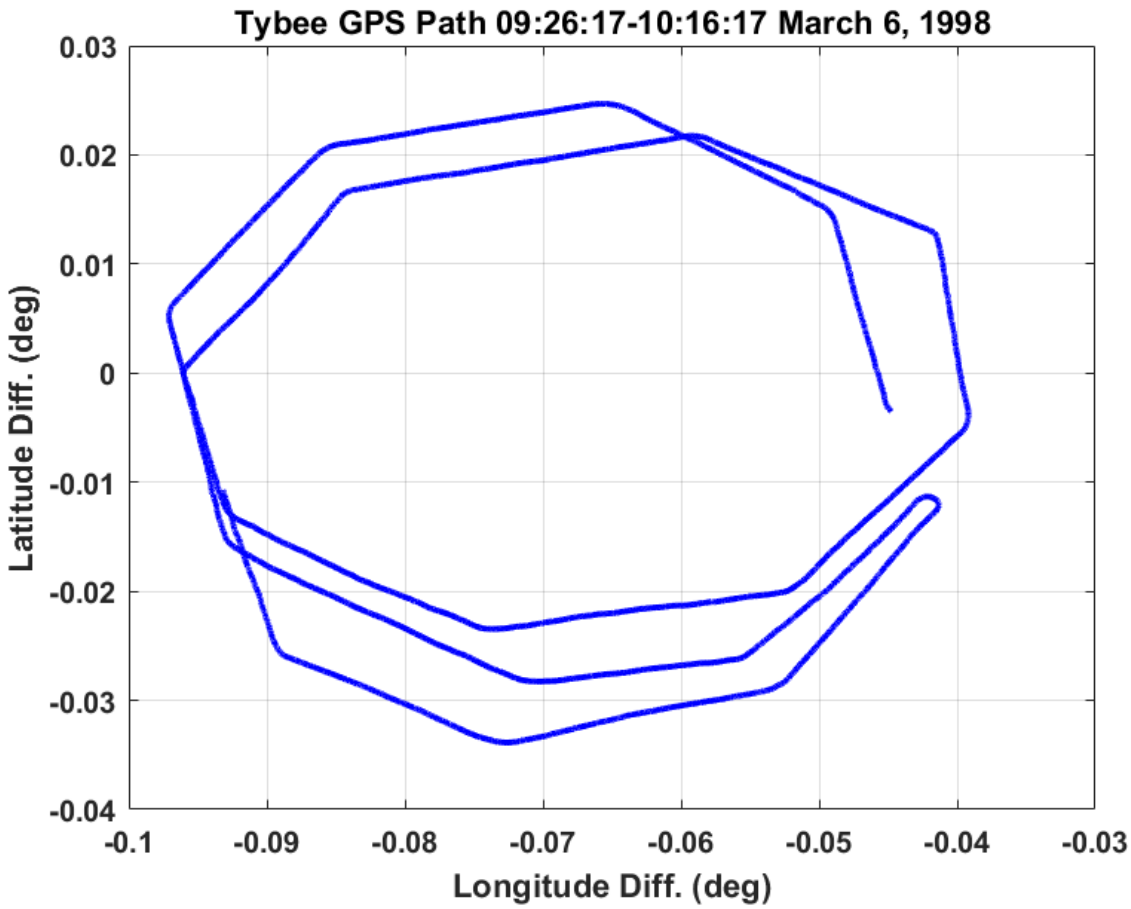
1. The two estimates are correlated.
2. They have similar variances.
3. Both estimates have a similar number of negative values, which, according to the Map-Drift formula, should not happen if the ship's path is straight. [R*A = u^2, with a stationary platform and no tracker]

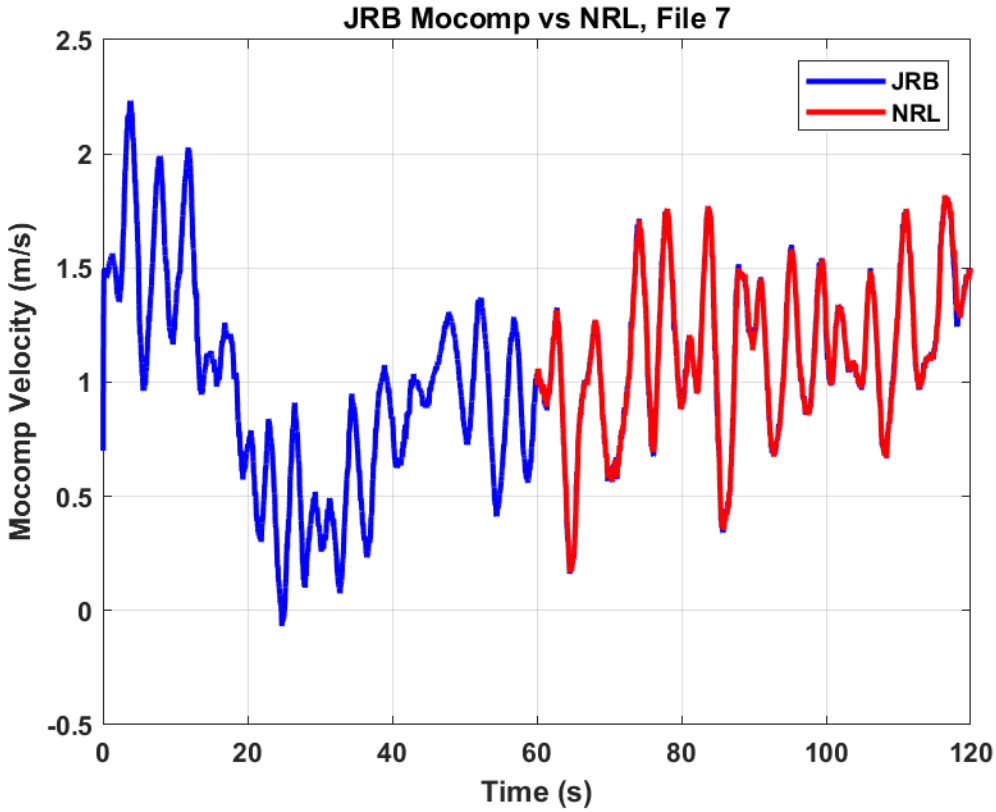
My interpretation is that there is a common cause of this variation. In fact, my candidate for the cause is simple – I think that the helmsman cannot steer an accurate enough course. The spatial deviation required to explain the errors in acceleration is in the range 0-5 meters, with a standard deviation of 2 meters.

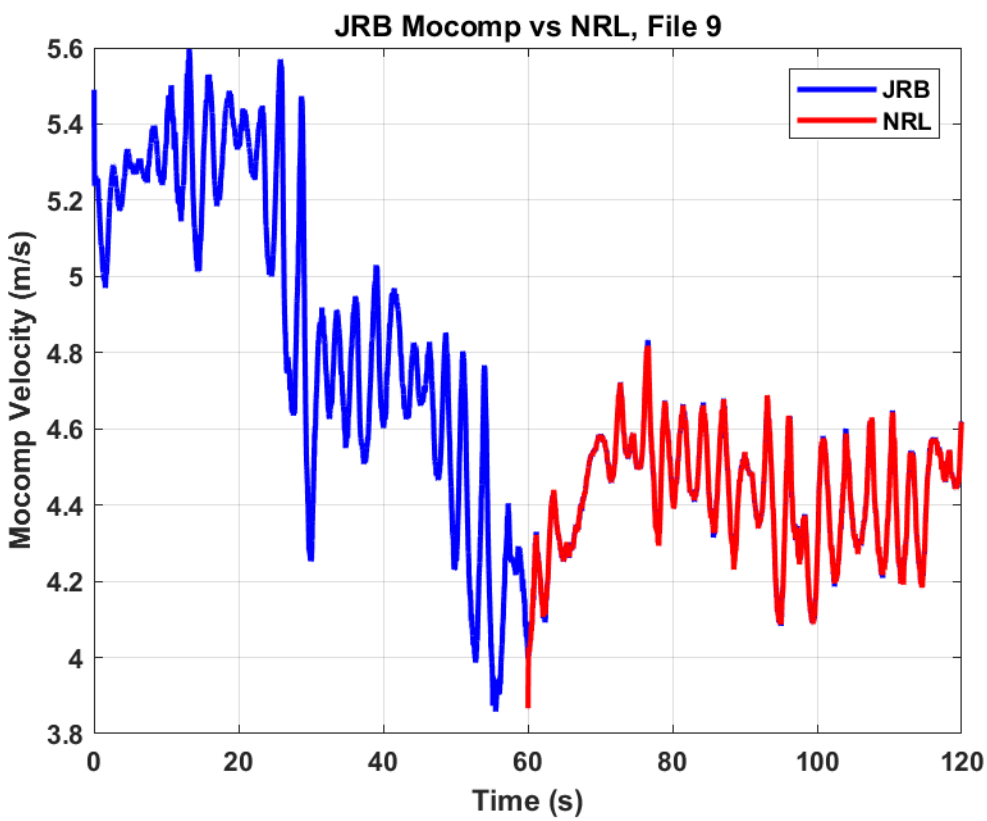
The second chart above plots the accelerations for the GPS and ISAR data for the 60-second calculations against the true value calculated from the mean cross-range velocities from the GPS data. I call this the true value since the mean range component of the GPS velocity agrees with the ISAR data and the mean component of the cross-range component must have similar errors since the GPS data is just a set of X-Y points. There is clearly a lot of error in both the GPS and ISAR estimates. However, the two estimates are clearly correlated. My conclusion from these facts is that the two methods are measuring real accelerations. In other words, the Tybee is not executing a perfectly straight course but instead is trying to do so but is suffering steering variations to the intended course.

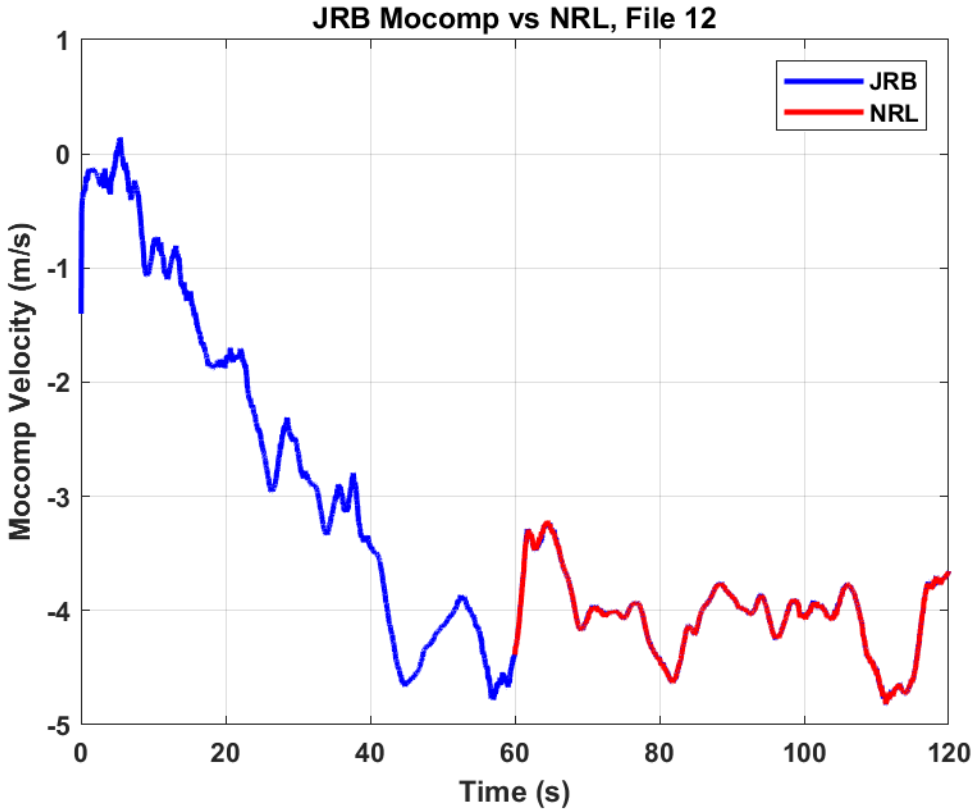


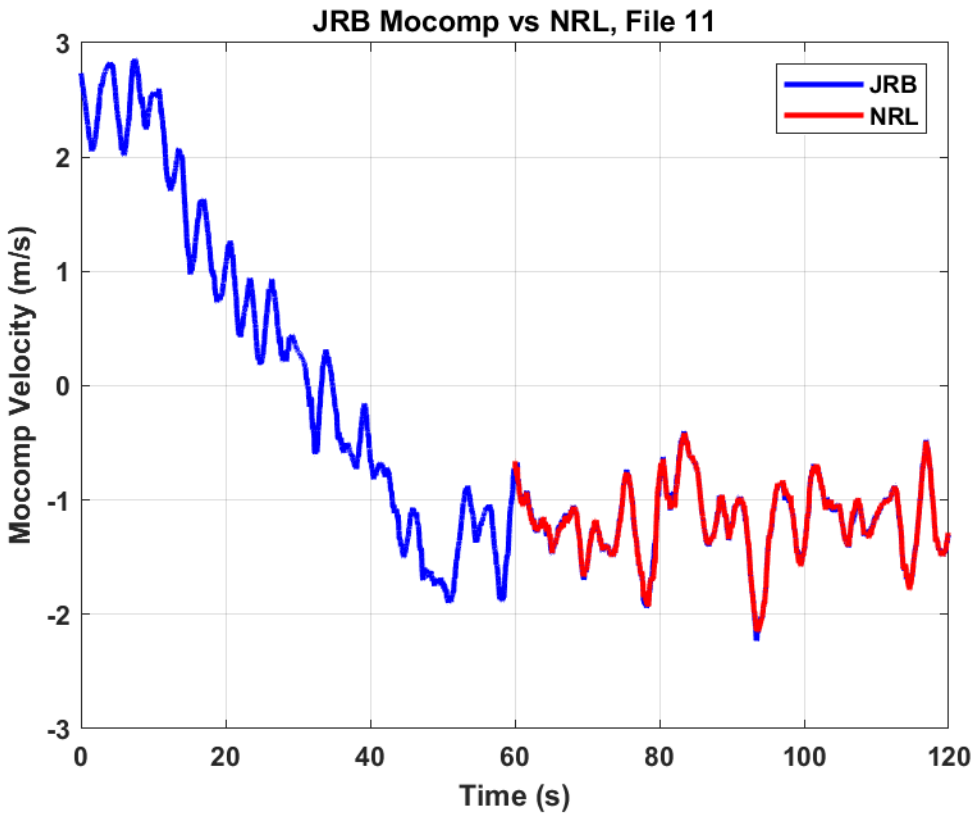

Tybee GPS Path 09:26:17-10:16:17 March 6, 1998
0.025
0.02
0.015
0.01
0.005
0
-0.005
-0.01
-0.015
-0.02
-0.025
Latitude Diff. (deg)
-0.1
-0.09
-0.08
-0.07
-0.06
-0.05
-0.04
-0.03
Longitude Diff. (deg)


Tybee GPS Path 09:26:17-10:16:17 March 6, 1998
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
-0.04
Latitude Diff. (deg)
-0.1
-0.09
-0.08
-0.07
-0.06
-0.05
-0.04
-0.03
Longitude Diff. (deg)


JRB Mocomp vs NRL, File 7
JRB
NRL
Mocomp Velocity (m/s)
Time (s)
2.5
2
1.5
1
0.5
0
-0.5
0
20
40
60
80
100
120


JRB Mocomp vs NRL, File 9
JRB
NRL
Mocomp Velocity (m/s)
Time (s)
5.6
5.4
5.2
5
4.8
4.6
4.4
4.2
4
3.8
0
20
40
60
80
100
120


JRB Mocomp vs NRL, File 12
JRB
NRL
Mocomp Velocity (m/s)
Time (s)
1
0
-1
-2
-3
-4
-5
0
20
40
60
80
100
120


JRB Mocomp vs NRL, File 11
JRB
NRL
Mocomp Velocity (m/s)
Time (s)
3
2
1
0
-1
-2
-3
0
20
40
60
80
100
120